

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/248879144>

Residual Tensile Properties of GFRP Reinforcing Bars after Loading in Severe Environments

Article in *Journal of Composites for Construction* · October 2006

DOI: 10.1061/(ASCE)1090-0268(2006)10:5(370)

CITATIONS

66

READS

436

4 authors:



Ahmed Debaiky

Benha University

35 PUBLICATIONS 549 CITATIONS

[SEE PROFILE](#)



Brahim Benmokrane

Université de Sherbrooke

667 PUBLICATIONS 20,064 CITATIONS

[SEE PROFILE](#)



G. Nkurunziza

Institut des Sciences Agronomiques du Burundi

7 PUBLICATIONS 191 CITATIONS

[SEE PROFILE](#)



Patrice Cousin

Université de Sherbrooke

61 PUBLICATIONS 1,698 CITATIONS

[SEE PROFILE](#)

Residual Tensile Properties of GFRP Reinforcing Bars after Loading in Severe Environments

by

Ahmed S. Debaiky¹, Gilbert Nkurunziza², Brahim Benmokrane³, and Patrice Cousin⁴

¹ Postdoctoral Fellow, Department of Civil Engineering, Faculty of Engineering, University of Sherbrooke, Sherbrooke, Quebec, Canada, J1K 2R1

² Formerly PhD student, Department of Civil Engineering, Faculty of Engineering, University of Sherbrooke; Project Manager, Les Laboratoires de Shermont, Teknika HBA Inc., 2605 Bonin, Sherbrooke, Quebec, Canada, J1K, 1C5

³ NSERC Research Chair Professor in FRP Reinforcement for Concrete Structures, Department of Civil Engineering, Faculty of Engineering, University of Sherbrooke, Sherbrooke, Quebec, Canada, J1K 2R1

⁴ Post-doctoral fellow, Department of Civil Engineering, Faculty of Engineering, University of Sherbrooke, Sherbrooke, Quebec, Canada, J1K 2R1

³ Corresponding Author

Phone: (819) 821-7758

Fax: (819)-821-7974

Brahim.Benmokrane@USherbrooke.ca

Manuscript submitted for revision and possible publication in

ASCE Journal of Composites for Construction

Revised December 2005

ABSTRACT

The long-term behavior of glass fiber-reinforced polymer (GFRP) reinforcing bars is one of the most critical issues for the acceptance of these materials as reinforcement for concrete structures. There is a high demand for experimental studies to investigate the stability of the tensile strength, ultimate elongation, and elasticity modulus. GFRP reinforcing bars inherently have a low elasticity modulus, which must not significantly decrease over time under loading or the serviceability behavior of the concrete element containing them will be jeopardized. This paper evaluates the residual tensile properties of three sizes of sand-coated GFRP reinforcing bars in alkaline and water environments combined with sustained loading and elevated temperature. Bar diameters of 15.9 (#5), 12.7 (#4), and 9.5 mm (#3) were loaded for different durations, then tested in axial tension for residual tensile properties. The test periods varied from 1 to 4 months under elevated temperature to hasten degradation and simulate extended service periods. The reduction in tensile strength was found to be 7% to 13% of the guaranteed strength for the three bar sizes under elevated temperature, which is at least 26% higher than the specified design strength as recommended by ACI 440.1R-03. More importantly, no significant change in the elastic modulus was observed.

INTRODUCTION

During the last decade, there has been an important increase in the use of Fiber Reinforced Polymer (FRP) composite bars as concrete reinforcement in the construction industry, mainly because of their non-corrosive nature. Other attractive properties include lightweight, high strength, and stiffness, and ultraviolet resistance. There are two primary types of FRP reinforcing bars: glass (GFRP) and carbon (CFRP). GFRP is gaining more popularity as tensile reinforcement or an alternative to traditional steel reinforcement in bridge deck slabs, mainly due to its significant lower cost compared to CFRP. Several GFRP fiber types are available, such as High Strength (S), Chemical Stable (C), Alkali-Resistance (AR), Electrical grade (E), and Advantex™, in addition to the traditional E-Glass fibers used in this research. Extensive research in North America, Europe, and Japan has established the mechanical abilities of GFRP bars. The lack of long-term durability data, however, is cited as a primary obstacle to wider acceptance of FRP materials in infrastructure applications (Harries et al. 2003). The current design guidelines and codes (ACI 440.01R-03, CAN/CSA-S806-02, ISIS-M03-01, CAN/CSA-S6-00, JSCE-1997) specify different strength reduction factors to account for the effects of sustained stress, fatigue, and environmental conditions. The Canadian guidelines allow higher values but the use of certain FRP types is restricted in some conditions (Byars et al. 2001). The development of environmental reduction factors (C_E) was an initial solution to overcome the lack of durability data and to allow design engineers to use the new materials sooner. Several design guidelines and codes currently use conservative environmental reduction factors as low as 0.7 in addition to the standard load and resistance factors. When environmental effects are combined with sustained loading factors, the total reduction factor recommended by ACI 440.1R-03 is 0.14-0.16. This severe reduction diminishes the advantages of the material's high initial strength.

1 The research focusing on the durability of FRP materials is considered at an early stage,
2 especially given the wide spectrum of durability issues and effects to be investigated;
3 Benmokrane and Rahman gathered and published early efforts on durability research in 1998. At
4 that time, most of the research focused on evaluating the durability of the material under
5 different environments with no loading. The load effect has been identified as a major
6 contributor in the response of the FRP to the harsh environments, because the load causes
7 microcracks in the resin bonding the fibers; these cracks give greater access for the environment
8 to attack the fibers. At that time, the authors concluded that any durability study must be
9 preformed under sustained loading to better resemble field conditions.

10 The developments in the materials used, manufacturing technique, and quality control have been
11 proven to have a serious impact on the durability of FRP products (Tannous and Saadatmanesh,
12 1998).

13 The loss of structural integrity of the glass fiber in alkaline environment is primarily due to the
14 attack of OH^- ions (Gerritse, 1992). The early generation of E-glass/Vinylesters bars suffered
15 loss in strength up to 70% when exposed to alkaline solution for 9 month under only 10% of
16 their nominal strength (Sen et al. 2002). Those tested were produced in the early 1990s and were
17 used as tensile reinforcement in a bridge pier in San Diego, California in 1993. These bars failed
18 within 25 days when exposed to an alkaline solution (pH 13.5). Similar observations were
19 reported by Micelli and Nanni in 2004. The newer generations have shown much higher
20 durability in alkaline solutions (pH 12.5) under 65% stress level, failing after 83 to 248 days
21 (Mutsuyoshi, 2001). This did not hold true for glass/Epoxy bars (Benmokrane et al. 1998). E-
22 glass/Vinylester bars have shown similar integrity when immersed in water at 60° C for 434 days,
23 losing 44% of their bending strength (Nishizaki and Meiarashi, 2002). Matrix type plays a major

1 role in protecting the glass fibers from alkaline attack; recent findings suggest that Vinylester
2 exhibits lower diffusivity and provides better protection than polyester (Tannous and
3 Saadatmanesh, 1999).

4 It should be noted here that the use of alkaline solution in the durability studies aims at
5 accelerating the damage mechanism. The bars in the field come in contact with concrete in a
6 solid state, which causes much slower effect than the solution used. Alkaline solution, on the
7 other hand, naturally migrates through microcracks into the bar. Recently, GFRP used as flexure
8 reinforcement in concrete beams subjected to very harsh conditions for 120 days have shown
9 considerable improvement in durability with strength loss of only 5% to 30% (Almusallam et al.,
10 Alsayed et al. 2002). The beams were conditioned in tap or seawater at 40° C temperature and a
11 stress level of 25% of their ultimate strength. The long-term degradation of the GFRP reinforcing
12 bars having a polymer with a glass transition temperature around 120°C (such as those used in
13 this study) in an alkaline environment can be evaluated experimentally in relatively short periods
14 using heat acceleration at temperatures up to 60°C (Benmokrane et al. 2002). This level of
15 temperature should significantly accelerate the degradation mechanism without jeopardizing the
16 material itself and has been used recently in several studies (Kajorncheappunngam et al. 2002,
17 Nishizaki and Meiarashi, 2002, Tannous and Saadatmanesh, 1999).

18 The main objective of the reported study was to evaluate certain long-term characteristic of FRP
19 reinforcing bars in terms of residual strength and elastic modulus under different environmental
20 conditions. The following specific objectives were targeted:

- 21 – Assess the residual tensile properties of GFRP reinforcing bars in alkaline or water media, at
22 elevated or ambient temperature, combined with sustained loading.
- 23 – Compare the residual tensile properties to the design limits given by ACI 440.1R-03.

RESEARCH SIGNIFICANCE

There is an urgent demand for durability data in the field of FRP composites for civil infrastructures. This research focuses on evaluating the residual tensile properties of traditional E-glass/Vinylester FRP reinforcing bars under loading in common media of alkaline and water solution. Large-scale laboratory equipment was designed and built specifically for this purpose. The study included the combined effects of high temperature and alkaline or water environments on GFRP bars under axial tension. The elevated temperatures were used to accelerate GFRP bar absorption of the medium solution. The data obtained focus on the residual strength and stiffness as affected by the environmental conditions. This research provides new data on the durability of the newest generation of reinforcing GFRP bars, which is the main reason behind harsh strength reduction factor currently adopted by most design code mainly due to the lack of durability data. The bars were tested under the combined effect of sustained load, elevated temperature, and aggressive surrounding media. The study proves that almost no change in the elastic modulus of the GFRP bars will take place; lower drops in the ultimate strength compared to the available literature were observed. The findings suggest the need to adopt less conservative design limits than those currently used.

FLEXURAL DESIGN USING GFRP BARS

The flexural design of beams and slabs using conventional steel reinforcement usually limits the stress in the rebar to 60% of the yield stress under unfactored service loads. This restricts the strain in the reinforcement to about $1200 \mu\epsilon$ and the crack width in the concrete to about 0.4-0.5 mm. On the other hand, while the GFRP reinforcement bars have an ultimate strength of between 600 and 800 MPa (1.5 to 2 times the yield stress of steel), their low modulus of elasticity (0.2

times that of steel) dictates the use of much more stringent strain levels to achieve an acceptable crack width of 0.7 mm (since GFRP bars are non-metallic and no risk of corrosion from salt penetration through cracks exists). ISIS-M03-01 guidelines choose to limit the strain level for GFRP bars to 2000 $\mu\epsilon$ under sustained loading.

Limits on the allowable stress in GFRP bars are enforced by ACI-440.1R-03 to avoid creep rupture of the bars under sustained or fatigue loads. For fatigue regimes, a maximum of only 0.2 of the design strength is permitted (i.e. 0.14 of the guaranteed strength of the bars, since the design strength is 0.7 of the guaranteed strength as shown in Table 3). The overall reduction factor in the Canadian standard CAN/CSA-S806-02 is less restrictive than other codes, allowing a maximum stress of 0.3 of the material strength applied to service loads. Canadian Highway Bridge Design code (CAN/CSA-S6-00) reduces the specified strength of GFRP reinforcing bars by 0.6 (0.75x0.8) under factored loads. Table 1 summarizes the allowable stresses given by the different design codes compared to the values used in this research. As shown, the stresses used in this research were higher than the service design stress limits to evaluate the durability of the bars beyond the current code limits.

EXPERIMENTAL PROGRAM

Parameters Included

Three main parameters were used in this research:

- i- *Sustained tensile stress*: Axial tension was applied on all bars during the entire duration of the test to mimic field conditions. The stress level was kept at 19% to 29% of the guaranteed tensile stress, corresponding to tensile strains of 2000-4600 $\mu\epsilon$, depending on bar size. These levels of strains are about 1.4 to 3.3 times the values recommended by

ACI440.1R-03 for creep rupture (20% of the design strain, equivalent to 1400 $\mu\epsilon$ on average). This was done to explore the material's potentials and evaluate how conservative the current codes and guidelines are. The sustained tensile stress is a major factor in this durability study as it promotes the ability of external media to migrate through the micro-cracks in the polymer matrix around the fibers, increasing the attack level on the glass fiber.

ii- *Surrounding media*: Two different media were used to simulate the most common state of the reinforcing bars in the field: deionized water and alkaline solution. The alkaline solution (pH = 12.9), which simulates the cement paste, was kept at a constant pH throughout the test. It is critical to realize that the alkaline solution used was liquid, which produces higher effect on GFRP bars than concrete.

iii- *Temperature*: Both ambient and elevated temperatures were used, with ambient temperature being 20° C and elevated temperature 45 to 73° C. Table 2 gives the equivalent years of actual field conditions for laboratory test periods at elevated temperature based on a model generated from data by Vijay and GangaRao (1999):

$$\text{Age in calendar days / Days of chamber conditioning} = 0.098 \cdot e^{0.0558T}$$

where T = temperature in °F

The Canadian average daily temperature is much lower than that used to build the model, so the actual equivalent field life should be greater than those given in the table. It should be noted here that several models were developed by other researchers, and they may yield different correlation between laboratory exposure conditions and service life. The above relation has been adopted by many others because of its simplicity and relative accuracy.

1 The surrounding media and the temperature were used to create four different exposure
2 combinations. The sustained stress was a common parameter with all specimens.

4 **Materials**

5 The GFRP reinforcing bars used in this research are the second generation of GFRP produced by
6 Pultrall Inc. (Thetford Mines, Quebec, Canada). These GFRP bars are manufactured by
7 combining the pultrusion process with an in-line coating process for the outside sand surface
8 (Pultrall Inc.). The GFRP bar is made from high-strength E-glass fibers (75% fibre by volume)
9 with a Vinylester resin, additives, and fillers. Table 3 gives the main tensile properties of the bars
10 in terms of average ultimate strength, modulus of elasticity, and average ultimate strain. The
11 tensile properties of the bars were determined on representative reference specimens (non-
12 conditioned) in accordance with ACI 440.3R-04. The guaranteed and design tensile strengths are
13 also given, based on ACI 440.1R.03.

14 The GFRP bars were tested while embedded in either deionized water with a pH value of
15 approximately 7, simulating 100% relative humidity, or an alkaline solution simulating the
16 concrete environment. The solution was composed of 1185 g of Ca(OH)_2 , 9 g of NaOH, and 42 g
17 of KOH in 10 L of deionized water according to ACI 440.3R-04 recommendations. The pH of
18 this solution was 12.8 as recommended by ACI 440.3R-04. The pH value was periodically
19 checked and kept constant during all the time of exposure by adding additional compounds of by
20 total replacement.

Test Specimen

The GFRP bars used in this study were cut to a length of 1200 mm; each end was placed in a 410-mm-long steel tube grouted with a resin-mortar matrix to create grip ends for axial tension, as shown in Figure 1a. The middle 480 mm of the specimen was instrumented with electrical strain gages (ESG) to monitor longitudinal strains. In order to expose the specimens to the specific solution, the middle part of the bar was inserted into a plastic tube measuring 63 mm in diameter and 250 mm long, which served as a reservoir. One end of the tube is sealed with an adhesive resin, then the tube is filled with the solution to simulate the selected type of exposure and the other end sealed. The reservoir is equipped with a flexible plastic tube 5 mm in diameter to allow for frequent replacement of the solution. Figure 1b provides a close-up view of typical specimens ready for installation. A thermocouple wire is placed inside the reservoir for temperature measurement, along with the ESGs.

Experimental Set-up and Test Procedure

The test setup consists of several self-reacting, load-magnifying steel frames and a heating tunnel system. The test specimens were fastened in the steel frames from both ends, where axial tension was applied. The desired tension force was obtained by putting gravel onto the lever arms (Figure 2). The sustained load was kept constant for the duration of the test. The steel frames were positioned to put the middle part of the test specimens inside the heating tunnel (Figure 3). The temperature was automatically controlled and monitored. At the end of each conditioning phase, the specimen was removed and physically examined before being tested to failure in axial tension to measure the effect of environmental conditioning on mechanical and physical properties. Properties such as glass transition temperature (T_g) were also evaluated on reference and conditioned specimens using the differential scanning calorimeter (DSC).

Since solution exposure results in degradation of the GFRP bars due to solution absorption, elevated temperature was used to accelerate the absorption rate. The elevated temperature in this testing program varied from 45 to 73°C, while the tensile sustained load varied from 18 to 29% of the guaranteed short-term tensile strength. The temperature variation resulted from endless difficulties in controlling the temperature inside the entire tunnel, the actual temperatures in every specimen was monitored and recorded over test period using thermocouple. For most of the test period, the recorded temperatures were $\pm 5^{\circ}\text{C}$ from the target value with rare occasions of high variability of up to $\pm 13^{\circ}\text{C}$. The summary of the test durations is presented in Table 4.

Microstructural Analysis

Microstructural analysis were performed to study the effect of conditioning on the physico-chemical structure of the GFRP reinforcing bars. Three techniques have been used in the present research:

1. Scanning Electronic Microscopy (SEM) to study the structure and degradation (delamination, microcracking, corrosion) of the GFRP materials (fibres, resin matrix, and interface glass fibre/resin).
2. Differential Scanning Calorimetry (DSC) to evaluate the thermal properties of the polymeric resin (glass transition temperature, post-curing, plasticization due to water).
3. Fourier Transform Infrared Spectroscopy (FTIR) to measure the chemical degradation of the vinylester resin, i.e. the hydrolysis reaction caused by water and alkalis.

Scanning Electronic Microscopy (SEM) results were obtained with a Jeol microscope. The samples were placed in epoxy resin and cut with low speed saw and polished with diamond pastes. DSC measurements were obtained using nitrogen for temperatures up to 200°C at a heating rate of 10°C/min. The samples (20-30 mg) were cut from the core of fractured GFRP

bars. Fourier Transform Infrared (FTIR) analysis was conducted using spectrometer with a resolution of 2cm^{-1} . Dry GFRP bar samples (20 g) were crumbled using a motorized grinder, and then a small amount of powder (3-5 mg) was mixed with potassium bromide (IR Grade), and pressed to obtain pellets ready for analysis.

TEST RESULTS

Tables 5 to 7 list the results obtained for bar sizes 16, 12.7, and 9.5 mm, respectively. The results include the residual tensile stress and strain, and their relations to the code limits and the residual elastic modulus. The following discussion sheds some light on these results and their significance. The long-term creep strain versus time (417 days) of the 9.5 mm bars under sustained and environment are published elsewhere (Nkurunziza et al. 2005).

Residual Ultimate Strain

The residual strains of the different bar sizes tested in this research program are compared to each other and to the original value. Only the results from bars tested under high temperature will be discussed, since high temperature was a common factor used with all bar sizes and is of greater importance.

The largest bar tested (16 mm in diameter) at high temperature for 1 and 2 months recorded average residual strains of 1.35% and 1.31%, respectively, in water and 1.37% and 1.23% in alkaline solution. While the two-month period at high temperature is equivalent to 39 years of field exposure (according to Vijay and GangaRao), the lowest value obtained (1.23%) is 69% higher than the design value suggested by ACI 440. In the case of the 12.7-mm bars, the residual strain after 3 months at high temperature was 1.22% and 1.15% in water and in alkaline solution, respectively. The lower strains as compared with those in the larger bar depend on both diameter

and duration. The smallest bar tested (9.5 mm) over 2 months at elevated temperature (equivalent to 40 years of field exposure) recorded residual strains of 1.70% and 1.62% in water and alkaline solution, respectively, which correspond to 2.43 and 2.31 times the design strength considering the original moduli of elasticity given in Table 1. It should be noted that the original ultimate strains for the 16-, 12.7- and 9.5-mm bars are 1.4%, 1.5%, and 1.6%, respectively. The high residual strain in the 9.5-mm bar can be attributed to higher curing of the matrix as a result of the elevated temperature, and to lower number of micro-structural defects.

In general, all the bars tested exhibited very satisfactory residual strain values after exposure to high temperature and sustained tensile loading for up to 4 months.

Residual Tensile Strength

Column 5 in Tables 5 to 7 presents the experimental residual tensile strength results for the entire test after exposure. The ratios between the individual residual tensile strength values and the short-term ultimate, guaranteed, and design tensile strengths are also presented (Figure 4). Since the three bar sizes tested have significantly different tensile strengths, it is more appropriate to discuss the results in terms of ratios of residual to original strengths, rather than absolute values.

The guaranteed strength f_{fu}^* , and design strength f_{fu} specified by ACI 440 is used for this purpose.

After one month at elevated temperature, the 16-mm bars had 104% and 106% of their original f_{fu}^* in water and alkaline solution, respectively (148% and 151% of f_{fu}). These values changed to 97% and 91% after two months (139% and 130% of f_{fu}), which means that the bars well exceeded design requirements.

The 12.7-mm bar exhibited a slightly better performance after three months at elevated temperature with 93% and 95% of the original f_{fu}^* in water and alkaline solution, respectively (138% and 135% of f_{fu}). It is believed that the smaller bar size yields greater curing of the materials under the elevated temperature used, hence causing less strength loss. Lastly, the 9.5-mm bar returned the same range of results after 2 months of loading at elevated temperature.. The residual strengths were 110% and 101% of the original f_{fu}^* in water and alkaline solution, respectively (157% and 144% of f_{fu}). The extended duration appears to have had no significant harmful effect on the bar, since the design strength was well exceeded and despite the harsh media used.

Effect on Elastic Modulus

Results for all bar sizes are given in the right columns of Tables 5 to 7 and illustrated in Figure 5. All values reported were found using LVDT readings over the middle 200-mm of the bars. Experimental results indicate that the residual modulus of elasticity (E_{res}) was not much different from the original values for all bar sizes and durations. For the 16- and 12.7-mm bars having an original modulus of 42 GPa, the residual modulus ranged from 39.6 GPa (for the 16-mm bar in alkaline solution for 2 months) to 46.3 GPa (for the 12.7-mm bar in alkaline solution for 3 months). Very few specimens displayed a drop in elastic modulus (mostly in alkaline solution). The 39.6-GPa average represents only a loss of 5.7% in elastic modulus, which is a very encouraging ratio. The 9.5-mm bars, with an original modulus of 40 GPa, recorded average residual moduli of 38.1 GPa and 37.1 GPa in water and alkaline solution, respectively, after 2 months of exposure to

1 elevated temperature under load. An examination of the individual bar results reveals that very
2 few specimens had lower elastic moduli (the lowest was 35.4 GPa).

3 The results of the residual elastic modulus suggest that the strength and strain decrease at almost
4 the same rate, keeping the elastic modulus. A more detailed discussion into this issue is furnished
5 in another publication based on creep monitoring of the same GFRP bars (Nkurunziza et al.,
6 2005).

7 **Microstructural Analysis**

8 Degradation may affect the physico-chemical structure of FRP composite materials in different
9 ways: the polymeric matrix, the fibres and/or the interface between these two components can be
10 corroded. The hydrolysis or the formation of microcracks in the resin, the dissolution of glass
11 fibres, and the debonding at the interface of fibres/matrix are the most common phenomena
12 occurring in a GFRP material submitted to a corrosive environment. To have a better
13 understanding of the causes behind the strength loss, SEM, FTIR, and DSC analysis have
14 therefore been conducted on reference and conditioned GFRP samples.

15 ***Scanning Electronic Microscopy (SEM) Analysis:***

16 SEM analysis has been used to study the physical structure of the material by analyzing cross
17 sections of untested GFRP bars and rupture areas of tested bars. Figure 6 shows the cross
18 sections of unconditioned and conditioned samples. Micrograph 6-A shows the fibres located
19 just under the sand/resin coating (the GFRP bar used in this work are sand-coated, (Pultrall, inc.).
20 Examination of the fiber/matrix interface shown in micrograph 6-B shows no visible voids
21 (debonding) between the fibers and the vinyl ester resin. However, the different conditionings
22 did affect the interface only in the region located just below the external surface of the bar.
23 Micrographs 6-C and 6-D show a typical state of the interface of samples conditioned in water

1 and alkaline solution, respectively. A very thin void can be observed around the fibre on
2 micrograph 6-C. The thickness of the gap increases in samples conditioned in alkaline solutions
3 as seen on micrograph 6-D. In this case, debonding is clearly observed. These observations can
4 be made with several samples, which eliminate the possibility of them being produced during the
5 sample preparation steps such as cutting and polishing. In some cases, voids can be observed in
6 non conditioned samples due to manufacturing defects. However, delamination in the outer ring
7 of the bars just below the surface coating is more often detected in conditioned samples.
8 Moreover, the gap width between the fibres and the resin seems to increase with time,
9 temperature and the degree of corrosivity of the solution. Consequently, it can be concluded that
10 the debonding observed is not due to sample preparation but to the conditioning.

11 Fractured areas of reference and conditioned samples mechanically tested have also been
12 examined. Figure 7 illustrates the delamination phenomenon of fibres at the rupture zone.
13 Micrographs 7-A and 7-B show a glass fibre embedded into the resin, the adhesion between the
14 two materials is intimate since no free space is visible. Moreover, the fracture did not occur at
15 the interface only but inside the resin matrix itself, which can be observed by the “torn”
16 appearance of the resin. When the degree of adhesion decreases, a free space starts to appear.
17 The top right part of micrograph 7-C shows the debonding area of a fibre which is still relatively
18 bonded to the resin. As delamination increases, the fibre surface area bonded to the resin
19 decreases (micrographs D and E). In this case, the fracture only occurs at the interface and the
20 fracture surface is mostly constituted of fibres only. Finally, fully debonded fibres appear as
21 shown in micrograph 7-F. The different steps of delamination can be observed in all the samples,
22 including reference samples. The different conditionings decrease the ratio of bonded/debonded
23 fibers, which cannot be determined by SEM analysis. It has to be noted that no sign of

degradation of glass fibres was detected in any conditioned sample analyzed. The surface of the fibers remains smooth even after a long conditioning period at high temperatures ($>60^{\circ}\text{C}$), which shows that the glass fibres are not affected by the alkaline medium reported in this study.

Differential Scanning Calorimeter (DSC) Analysis:

The mechanical properties of polymer-based materials are susceptible to thermal conditions. One of the most important thermal parameter is the glass transition temperature (T_g), which is related to the viscoelastic properties of the material and is defined as the temperature where a polymer in a vitreous state becomes viscoelastic. At this temperature, mechanical properties decrease because of the polymeric phase softening. The degradation may affect the resin and decrease T_g if polymeric chains are ruptured, such as through a hydrolysis reaction occurring in the presence of alkalis. The presence of water may also decrease T_g since it may have a plasticizing effect. In this study, T_g has been measured by DSC, which also detects the presence of post-curing phenomenon (also called post-polymerisation/crosslinking). This phenomenon occurs when a polymer has not been completely cured during the manufacturing process, leaving unreacted monomer in the material.

Table 8 shows T_g values for reference and conditioned GFRP specimens. The reference specimen has a greater T_g value in the second scan than the first. This shift indicates that the samples were not fully cured and that a post-curing phenomenon occurred during the second heating run. Nevertheless, the difference between the two T_g , and consequently the amount of unreacted resin, are small.

For specimens conditioned in deionized water at 60°C , the glass transition during the first scan occurred at a lower temperature than in the reference samples, whereas the second scan values

1 were similar to the reference values. This may be explained by the presence of water absorbed
2 during the conditioning period since water acts as a plasticizer and decreases the glass transition
3 temperature. The water evaporates when the sample is heated and its plasticizing effect
4 disappears, leading to the reference T_g in the second run.

5 The same phenomenon occurs with the specimens conditioned in the alkaline solution at 60°C.
6 The T_g of the second scan was not affected by the conditioning, which shows that the
7 degradation was reversible and the second scan yielded glass transition temperature close to the
8 reference T_g . As a general conclusion, any decrease in T_g is reversible and is due to the presence
9 of water in the matrix. No signs of chemical degradation are visible because when the
10 degradation of a resin, such as hydrolysis, induces chain rupture, the glass transition temperature,
11 T_g , is lowered since shorter chains are more flexible and need less energy to move than longer
12 chains. Then, the material becomes viscoelastic at a lower temperature. If T_g is unchanged, as
13 observed in the present case, it may be assessed that the resin is not chemically degraded.

14 ***Fourier Transform Infrared Spectroscopy (FTIR) Analysis***

15 The presence of alkaline ions may induce a degradation reaction of the backbone of the
16 polymeric chain of FRP composite resins containing ester units. This chemical reaction, called
17 hydrolysis, breaks the resin molecules, leading to the formation of hydroxyl groups such as
18 alcohols or carboxylic acids. It affects the structure of the polymer and weakens the material,
19 which will eventually be unable to play its role in the transfer of stresses between the fibres. The
20 best way to determine if a hydrolysis reaction occurs in the polymeric matrix is to use FTIR
21 technique. With this technique, the relative amount of hydroxyl groups in the sample is measured
22 by determining the ratio between the maximum of the band corresponding to the hydroxyl
23 groups at 3430 cm^{-1} and those of the band corresponding to the carbon-hydrogen groups at 2900

cm⁻¹, the C-H content being considered as constant. Figure 8 shows the spectra in the region of interest of 16 mm diameter samples unconditioned and conditioned in deionized water and the alkaline solution. The hydroxyl peak is located at the left of the spectrum and the C-H peak at the right. Since the vinyl ester resins and silica “naturally” contain hydroxyl groups, all the spectra present a strong absorbance band in the hydroxyl region. The ratios OH/CH are equal to 1.05, 1.30, and 1.45 at the maximum of absorbance for the reference sample and dry samples conditioned for two months at 60°C in water and alkaline solution, respectively. Conditioning in water increased the amount of hydroxyl units by 20%, whereas the sample conditioned in the alkaline solution contains 38% more hydroxyls than the reference sample. It may be concluded that a small hydrolysis reaction has occurred during the conditioning and that the presence of alkaline ions seems to enhance this reaction. It has also to be noted that, in the case of samples conditioned in alkaline solutions, the hydroxyl unit content could also provide from the hydroxyl anions which have diffused through the material into the core of the bar and not only from the hydrolysis reaction.

Mode of Failure

No change in the mode of failure was observed due to the extended exposure to the different conditioning agents used. When tested under axial tension, all bars failed by violent rupture of the fibers. The stress-strain relation for conditioned bars remained linear up to failure, the only changes being in the slope of the line (modulus of elasticity) and the ultimate stress/strain.

It should be noted here that the conditioned GFRP bars usually failed at the end section of the conditioning zone (edges of the plastic reservoir) as shown in Figure 9. A sudden change in bar properties occurs here, which led to failure initiation. Therefore, the resulting failure stresses stand as a lower limit and would have been higher had the entire bar been subjected to the

conditioning as in field situations. It can be concluded that the tested GFRP bars experienced acceptable levels of strength loss under the testing program's aggressive environmental conditions. The behavior of the tested products proves the advancements made in materials quality and manufacturing techniques in the last few years in comparison to earlier generations of GFRP bars.

Discussion

The ultimate strain of the GFRP reinforcing bars is originally low (1.4% to 1.8%), therefore the ultimate deformations and deflections of a GFRP-reinforced concrete structure will be smaller than those in a comparable steel-reinforced structure. Hence, engineers cannot afford any significant degradation in the ultimate strain of the GFRP bars with time. The FRP materials in general and glass type in particular have been generally criticized for their low modulus of elasticity. The low modulus is blamed for high deflection and other serviceability concerns. Therefore, it is very important to ensure that the already low modulus does not experience any significant deterioration due to extended exposure to harsh environments and sustained loading. The residual elastic modulus results are very positive and of significant importance to industry, since confidence in the long-term stability of the elastic modulus of the bars for serviceability control is a critical issue. Testing the GFRP bars for extended period of time under sustained loading has been proven to have no significant creep strain (Nkurunziza et al., 2005), these positive results should relieve some of the concerns the public and design engineers have regarding the long-term behaviour of the bars. It also will contribute, along with findings from other researchers, to introduce less stringent design limit in codes of practice in order to fully utilize the material capacity and reduce the cost of their use.

CONCLUSIONS

The residual tensile properties of GFRP reinforcing bars were investigated with respect to the combined effects of temperature and alkaline or water medium, in addition to constant sustained tension. Three different bar sizes (16, 12.7, and 9.5 mm) were tested at elevated temperature and ambient temperature for different lengths of time. The elevated temperature used for 2 and 4 months is believed to simulate field conditions of 40 and 80 years, respectively, based on the model of Vijay and GangaRao (1999). The alkaline and water media are representative of the most commonly encountered field conditions. The following conclusions were drawn from the study:

- A maximum of 11% reduction in tensile strength (compared to the guaranteed strength) was observed in the 9.5-mm GFRP bars after exposure to the combined effect of 60°C temperature and alkaline solution under sustained tensile stress equal to 29%.
- The residual tensile strength for the conditioned specimens was higher (a minimum of 130%) than the specified design strength recommended in ACI 440.1R-03 design guidelines.
- The reduction in the ultimate strain after conditioning falls within the design limitations. The lowest residual strain was still 43% higher than the specified design strain recommended by ACI 440.1R-03.
- No significant loss in elastic modulus was observed under the severe conditions used. This applies to the three sizes of bar tested. In addition, the smaller bar size showed a slight increase in modulus after conditioning. This may be due to more curing of the matrix as a result of the elevated temperature.

- 1 - The environmental reduction factors recommended by many codes and guides for the
2 allowable strength/strain of GFRP reinforcing bars are thought to be very conservative. They
3 may nevertheless be acceptable for a limited time due to the lack of research data.
- 4 - The microstructural analysis (SEM, DSC, and FTIR) show that the different conditionings
5 mainly induce a delamination/debonding mechanism, which weakens the glass fibre/resin
6 matrix interface and partly hinders the stress transfer between adjacent fibres. A moderate
7 chemical degradation due to the hydrolysis of the ester groups of the resin is also observed.
8 However, no degradation of the glass fibres was detected.

9

10 It is believed that the laboratory conditions are much more corrosive than the field conditions
11 since liquid media were used, unlike the hardened concrete in real structures. It is unlikely
12 that GFRP reinforcement materials will be subjected to these environments in service.
13 Consequently, the level of degradation observed in this study will not be reached and the
14 material strength should remain higher than in the present case.

16 **AKCNOWLEDGEMENT**

17 The authors are grateful to the Natural Science and Research Council of Canada (NSERC) and
18 the Network of Centers of Excellence ISIS Canada for financial support. The materials and
19 technical assistance from Pultrall Inc. were greatly appreciated. We would also like to thank the
20 technical staff at University of Sherbrooke for their dedication and thoroughness.

REFERENCES

1. Almusallam, T. H., AlSalloum, Y. A., and Alhozaimy, A. M., 2002, "Tensile Strength of GFRP Bars in Concrete Beams Under Sustained Loads at Different Environments", Proc., 2st Int. Conf. on Durability of Fibre Reinforced Polymer (FRP) Composites for Construction (CDCC 02), Montreal, PQ, Canada, May 29-31, pp. 523-534.
2. Alsayed, S. H., Alhozaimy, A. M., AlSalloum, Y. A., and Almusallam, T. H., 2002, "Durability of the New Generation of GFRP Rebars under Sever Environments", Proc., 2st Int. Conf. on Durability of Fibre Reinforced Polymer (FRP) Composites for Construction (CDCC 02), Montreal, PQ, Canada, May 29-31, pp. 651-663.
3. American Concrete Institute, ACI 440.1R-03, 2003, Guide for the Design and Construction of Concrete Reinforced with FRP Bars, American Concrete Institute, Farmington Hills, Michigan, 42p.
4. American Concrete Institute, ACI 440.3R-04, 2003, Guide Test Methods for Fiber Reinforced Polymers (FRPs) for Reinforcing or Strengthening Concrete Structures, American Concrete Institute, Farmington Hills, Michigan, 40p.
5. Benmokrane, B., Wang, P., Ton-That, T, Rahman, H., and Robert, J., 2002, "Durability of Glass Fibre Reinforced Polymer Reinforcing Bars in Concrete Environment", ASCE J. of Composite for Construction, V. 6, No. 2, Aug. pp. 143-153.
6. Benmokrane, B. and Rahman, A.H. (Eds.), 1998, 1st Int. Conf. on Durability of Fibre Reinforced Polymers (FRP) Composites for Construction, (CDCC 98), Sherbrooke, PQ, Canada, July 18-20.
7. Benmokrane, B., Rahman, A.H., Ton-That, M.-T and Robert, J.-F, 1998, "Improvement of the Durability Performance of FRP Reinforcement for Concrete Structures", Proc., 1st Int.

- Conf. on Durability of Fibre Reinforced Polymer (FRP) Composites for Construction (CDCC 98), Sherbrooke, PQ, Canada, July 18-20, pp. 571-598.
8. Byars, E. A., Waldron, P., DeJke, V, and Demis, S., 2001, “Durability of FRP in Concrete – Current Specifications and a New Approach”, FRP Composites in Civil Engineering, V. II, J.-G. Teng (Eds.), Elsevier Science Ltd., pp. 1497-1507.
 9. Canadian Standards Association, CAN/CSA-S6-00, 2000, Canadian Highway Bridge Design Code, CSA International, Rexdale, Toronto, Ontario, Canada, 734 p.
 10. Canadian Standards Association, CAN/CSA-S806-02, 2002, Design and Construction of Building Components with Fibre Reinforced Polymers, *Canadian Standards Association*, Rexdale, Ontario, 177p.
 11. Gerritse, A., 1992, “Durability Criteria for Non-Metallic Tendons in an Alkaline Environment”, Proc., 1st Int. Conf. on Advanced Composite Materials in Bridges and Structures, ACMBS-I, Neale, K. W., and Labossière, P., (Eds.), Sherbrooke, PQ, Canada, pp. 129-137.
 12. Harries, K. A., Porter, M. L., and Busel, J. P., 2003, “FRP Materials and Concrete – Research Needs”, Concrete International, V. 25, No.10, Oct. pp. 69-74.
 13. Intelligent Sensing for Innovative Structures, ISIS-M03-01, 2001, Reinforcing Concrete Structures with Fibre Reinforced Polymers, The Canadian Network of Centres of Excellence, ISIS Canada, University of Winnipeg, Manitoba, 81 p.
 14. Japanese Society of Civil Engineers, 1997, Recommendations for Design and Construction of Concrete Structures Using Continuous Fiber Reinforced Materials, Concrete Engineering Series, No. 23, 325 p.

15. Kajorncheappunngam, S., Gupta, R., and GangaRao H., 2002, "Effect of Aging Environment on Degradation of Glass-Reinforced Epoxy", ASCE J. of Composite for Construction, V. 6, No. 1, Feb. pp. 61-69.
16. Micelli, F. and Nanni, A., 2004, "Durability of FRP Rods for Concrete Structures", Construction and Building Materials Journal, V. 18, No. 7, pp. 491-503
17. Mutsuyoshi, H., Sumida, A. And Uomoto, T., 2001, "Alkali Resistance of Fibers, FRP Rods and Epoxy Resins", Proc., 5th Int. Conf. on Fibre-Reinforced Plastics for Reinforced Concrete Structures, FRPRCS-5, Burgoyne, C.J. (Ed.), Cambridge, U.K, Vol.1, July 14-16, pp. 479-513.
18. Nkurunziza, G., Benmokrane, B., Debaiky, A., and Masmoudi, R., 2005, "Effect of Sustained Load and Environment on Long-Term Tensile Properties of Glass FRP Reinforcing Bars", ACI Materials Journal, V. 102, No. 4, July-Aug. 2005, pp. 615-621.
19. Nishizaki, I, and Meiarashi, S., 2002, "Long-Term Deterioration of GFRP in Water and Moist Environment", ASCE J. of Composite for Construction, V. 6, No. 1, Feb. 2002, pp. 21-27.
20. Pultrall Inc, ISOROD Composite Reinforcing Rods, 2000, Technical Data Sheet, Pultrall Inc, Thetford Mines, PQ, Canada, 7 p.
21. Sen, R., Mullins, G., and Salem, T., 2002, "Durability of E-Glass/Vinylester Reinforcement in Alkaline Solution", ACI Struc. J., V. 99, No. 3, May-June 2002, pp. 369-375.
22. Tannous, F. E., and Saadatmanesh, H., 1999, "Durability of AR Glass Fiber Reinforced Plastic Bars", ASCE J. of Composite for Construction, V. 3, No. 1, Feb. 1999, pp. 12-19.

23. Tannous, F. R and Saadatmanesh, H., 1998, "Environmental Effects on the Mechanical Properties of E-Glass FRP Rebars", ACI Materials J., V. 95, No. 2, Mar-Apr 1998, pp. 87-100.
24. Vijay, P. V., and GangaRao, H. V. S., 1999, "Accelerated and Natural Weathering of Glass Fibre Reinforced Plastic Bars", Proc., 4th Int. Conf. on Fibre-Reinforced Plastics for Reinforced Concrete Structures, FRPRCS-4, Baltimore, USA, Nov. 1-4, pp. 605-614.

LIST OF TABLES

Table 1: Allowable stress limits given by selected design codes

Table 2: Age in natural (years) for different laboratory conditioning temperature and durations (based on Vijay and GangaRao model [20]).

Table 3: Tensile properties of the GFRP bar used in this study

Table 4: Summary of the durations of tests performed

Table 5: Results for the 16 mm bars.

Table 6: Results for the 12.7 mm bars (for four months).

Table 7: Results for the 9.5 mm bars

Table 8: Differential scanning calorimeter (DSC) analysis results

Table 1: Allowable stress limits given by selected design codes

Reference	Equation	Notes	Allowable stress (MPa)
ACI 440.1R-03	$0.20 f_{fu}$	f_{fu} : design strength including environmental reduction factor.	78
CAN/CSA-S6-00	$0.3 f_u$	f_u : failure strength (assumed guaranteed)	167
CAN/CSA-S806-02 (under factored loads)	$\phi_{FRP} \cdot F \cdot f_{pu}$	ϕ_{FRP} : material factor = 0.75 for GFRP F: factor depends on loads ratios, usual taken equal to 0.8 f_p : specified strength (guaranteed)	335
ISIS-M03-01	$\varepsilon = 0.002$	Stress-equivalent strain	84
Stress used in current research			100-162

Table 2: Age in natural (years) for different laboratory conditioning temperature and durations (based on Vijay and GangaRao model [20]).

$^{\circ}\text{C}$	30	40	50	60
$^{\circ}\text{F}$	86	104	122	140
1 month	1.0	2.7	7.3	19.9
2 months	2.0	5.3	14.6	39.8
4 months	3.9	10.7	29.1	79.5

Table 3: Tensile properties of the GFRP reinforcing bar used in this study

Property	Symbol	16 mm	12.7 mm	9.5 mm
Ultimate tensile strength, MPa	$f_{u,ave}$	580±15	639±27	658±10
Guaranteed tensile strength, MPa	f_{fu}^* ($f_{fu}^* = f_{u,ave} - 3\sigma$)	535	558	628
Design tensile strength (ACI 440), MPa	$f_{fu} = C_E \cdot f_{fu}^*$, $C_E = 0.7$	375	392	440
Modulus of elasticity, GPa	$E_{f,ave}$	42±2	42±2	40±2
Ultimate strain, %	$\varepsilon_{u,ave}$	1.4±0.12	1.5±0.12	1.6±0.2
Guaranteed strain, %	$\varepsilon_u^* = \varepsilon_{u,ave} - 3\sigma$	1.04	1.14	1.0
Design strain, %	$\varepsilon_u = C_E \cdot \varepsilon_u^*$	0.7	0.8	0.7

Table 4: Summary of the durations of tests performed

Bar size	Ambient temperature	Elevated Temperature
9.5 mm	14 months (417 days)	2 months
12.7 mm	1 months 4 months 8 months	4 months
16 mm	1 months 4 months 8 months	1 months 2 months

Every test was made using both alkaline and water mediums

Table 5: Results for the 16 mm bars.

	Series	Specimen I.D.	Temp-erature °C	Strength (tension test)				Strain (tension test)				Residual Elastic Modulus (GPa)
				Residual (MPa)	$\frac{f_{res}}{f_{u,ave}}\%$	$\frac{f_{res}^*}{f_{fu}^*},\%$	$\frac{f_{res}}{f_{fu}}\%$	Residual $\epsilon_{res},\%$	$\frac{\epsilon_{res}}{\epsilon_{u,ave}}$	$\frac{\epsilon_{res}^*}{\epsilon_u^*}$	$\frac{\epsilon_{res}}{\epsilon_u}$	
One Month	Water	16W-1M1	60	574	99	107	153	1.31	0.94	1.26	1.87	43.9
		16W-1M2	58	522	90	97	139	1.30	0.93	1.25	1.86	40.1
		16W-1M3	58	536	92	100	143	1.33	0.95	1.28	1.90	40.3
		16W-1M4	57	610	105	114	162	1.51	1.08	1.45	2.16	40.4
		16W-1M5	55	538	93	100	143	1.32	0.94	1.27	1.89	40.8
		Average	58	556	96	104	148	1.35	0.96	1.30	1.93	41.1
	Alkaline	16A-1M1	54	524	90	98	139	1.32	0.94	1.27	1.89	39.8
		16A-1M2	53	544	94	101	145	1.23	0.88	1.18	1.76	44.4
		16A-1M3	51	578	100	108	154	1.35	0.96	1.30	1.93	42.9
		16A-1M4	50	603	104	112	160	1.44	1.03	1.38	2.06	41.9
		16A-1M5	53	594	102	111	158	1.54	1.10	1.48	2.20	38.5
		Average	55	568	98	106	151	1.37	0.98	1.32	1.96	41.5
Two Months	Water	16W-2M1	58	533	92	99	142	1.36	0.97	1.31	1.94	39.2
		16W-2M2	61	516	89	96	137	1.23	0.88	1.18	1.76	42.1
		16W-2M3	60	484	83	90	129	1.25	0.89	1.20	1.79	38.6
		16W-2M4	60	538	93	100	143	1.34	0.96	1.29	1.91	40.3
		16W-2M5	60	538	93	100	143	1.35	0.96	1.30	1.93	39.8
		Average	60	522	90	97	139	1.31	0.94	1.26	1.87	40.0
	Alkaline	16A-2M1	61	518	89	96	138	1.38	0.99	1.33	1.97	37.7
		16A-2M2	61	454	78	84	121	1.11	0.79	1.07	1.59	40.7
		16A-2M3	60	519	90	97	138	1.30	0.93	1.25	1.86	39.8
		16A-2M4	61	477	82	89	127	1.20	0.86	1.15	1.71	39.8
		16A-2M5	60	471	81	88	125	1.18	0.84	1.13	1.69	40.0
		Average	61	488	84	91	130	1.23	0.88	1.18	1.76	39.6

Table 6: Results for the 12.7 mm bars (for four months).

Series	Specimen I.D.	Applied Stress, $\% f_{fu}^*$	Temperature $^{\circ}\text{C}$	Strength (tension test)				Strain (tension test)				Residual Elastic Modulus (GPa)
				Residual (MPa)	$\frac{f_{res}}{f_{u,ave}} \%$	$\frac{f_{res}}{f_{fu}^*} \%$	$\frac{f_{res}}{f_{fu}} \%$	Residual $\epsilon_{res} \%$	$\frac{\epsilon_{res}}{\epsilon_{u,ave}}$	$\frac{\epsilon_{res}}{\epsilon_u^*}$	$\frac{\epsilon_{res}}{\epsilon_u}$	
Water	13W-3M1	33.2	63	467	73	83	119	0.96	0.64	0.84	1.20	49
	13W-3M2	24.0	62	568	89	101	145	1.29	0.86	1.13	1.61	44
	13W-3M3	22.9	61	488	76	87	124	1.08	0.72	0.95	1.35	45
	13W-3M4	22.9	60	527	82	94	134	1.05	0.70	0.92	1.31	50
	13W-3M5	22.9	60	545	85	97	139	1.24	0.83	1.09	1.55	44
	13W-3M6	30.9	60	504	79	90	129	1.17	0.78	1.03	1.46	43
	13W-3M7	26.3	57	597	93	107	152	1.42	0.95	1.25	1.78	42
	13W-3M8	22.9	56	624	98	111	159	1.56	1.04	1.37	1.95	40
Alkaline	Average	25.8	60	540	85	93	138	1.22	0.81	1.07	1.53	44.6
	13A-3M1	24.0	62	479	75	86	122	1.17	0.78	1.03	1.46	41
	13A-3M2	20.6	61	520	81	93	133	1.16	0.77	1.02	1.45	45
	13A-3M3	22.9	61	545	85	97	139	1.11	0.74	0.97	1.39	49
	13A-3M4	27.5	51	534	84	95	136	1.11	0.74	0.97	1.39	48
	13A-3M5	25.2	49	548	86	98	140	1.03	0.69	0.90	1.29	53
	13A-3M6	22.9	46	555	87	99	142	1.23	0.82	1.08	1.54	45
	13A-3M7	27.5	45	524	82	94	134	1.22	0.81	1.07	1.53	43
	Average	24.4	57	529	83	95	135	1.15	0.77	1.01	1.44	46.3

Table 7: Results for the 9.5 mm bars

	Series	Specimen I.D.	Temperature °C	Strength (tension test)				Strain (tension test)				Residual Elastic Modulus (GPa)
				Residual (MPa)	$\frac{f_{res}}{f_{u,ave}}, \%$	$\frac{f_{res}}{f_{fu}^*}, \%$	$\frac{f_{res}}{f_{fu}}, \%$	Residual $\epsilon_{res}, \%$	$\frac{\epsilon_{res}}{\epsilon_{u,ave}}$	$\frac{\epsilon_{res}}{\epsilon_u^*}$	$\frac{\epsilon_{res}}{\epsilon_u}$	
Two Months	Water	9.5W-2M1	73	667	99	114	162	1.79	1.12	1.79	2.56	37.2
		9.5W-2M2	74	638	94	109	155	1.70	1.06	1.70	2.43	37.6
		9.5W-2M3	75	656	97	112	160	1.74	1.09	1.74	2.49	37.6
		9.5W-2M4	75	664	98	113	162	1.64	1.02	1.64	2.34	40.5
		9.5W-2M5	75	640	95	109	156	1.74	1.09	1.74	2.49	36.7
		9.5W-2M6	72	642	95	109	156	1.67	1.04	1.67	2.39	38.5
		9.5W-2M7	71	646	95	110	157	1.73	1.08	1.73	2.47	37.3
		9.5W-2M8	71	635	94	108	155	1.60	1.00	1.60	2.29	39.6
		9.5W-2M9	70	642	95	109	156	1.73	1.08	1.73	2.47	37.1
		9.5W-2M10	66	641	95	109	156	1.64	1.02	1.64	2.34	39.1
		Average	72	647	96	110	157	1.70	1.06	1.70	2.43	38.1
Alkaline	9.5A-2M1	74	500	74	85	122	1.41	0.88	1.41	2.01	35.4	
	9.5A-2M2	74	537	79	91	131	1.45	0.91	1.45	2.07	37.0	
	9.5A-2M3	73	594	88	101	145	1.58	0.98	1.58	2.26	37.7	
	9.5A-2M4	59	606	90	103	147	1.63	1.02	1.63	2.33	37.1	
	9.5A-2M5	58	628	93	107	153	1.76	1.10	1.76	2.51	35.6	
	9.5A-2M6	58	635	94	108	155	1.63	1.02	1.63	2.33	39.0	
	9.5A-2M7	56	648	96	110	158	1.71	1.07	1.71	2.44	37.8	
		Average	64	593	88	101	144	1.62	1.01	1.62	2.31	37.1
Fourteen Months (417 days)	Water	9.5W-14M1	Ambient	606	92.0	96.4	137.6	1.54	0.96	1.54	2.20	39.4
		9.5W-14M2	Temp.	587	89.2	93.4	133.4	1.37	0.86	1.37	1.96	42.7
		9.5W-14M3		587	89.3	93.5	133.5	1.52	0.95	1.52	2.16	38.8
		9.5W-14M4		649	98.7	103.4	147.6	1.55	0.97	1.55	2.21	42.0
		9.5W-14M5		621	94.3	98.8	141.0	1.48	0.93	1.48	2.12	41.8
		Average		610	92.7	97.1	138.6	1.49	0.93	1.49	2.13	40.9

	Alkaline	9.5A-14M1	539	82.0	85.9	122.6	1.32	0.83	1.32	1.89	40.8
		9.5A-14M2	605	91.9	96.3	137.5	1.44	0.90	1.44	2.05	42.1
		9.5A-14M3	584	88.8	93.0	132.8	1.38	0.86	1.38	1.97	42.3
		9.5A-14M4	637	96.8	101.4	144.7	1.56	0.98	1.56	2.23	40.8
		9.5A-14M5	412	62.6	65.6	93.6	1.01	0.63	1.01	1.45	40.6
		<i>Average</i>	555	84.4	88.5	126.2	1.35	0.84	1.35	1.92	41.3

Table 8: Differential scanning calorimeter (DSC) analysis results

Specimen	T_g (scan 1) (°C)	T_g (scan 2) (°C)
Reference	114	120
Conditioned (Deionized Water)	105	120
Conditioned (Alkaline Solution)	106	120

LIST OF FIGURES

Figure 1a: Details and dimensions of GFRP test specimen.

Figure 1b: GFRP test specimens ready for installation.

Figure 2: Sustained tensile forces applied on GFRP bars through self load-magnifying steel frames.

Figure 3: Steel frames positioned inside the heating tunnel.

Figure 4: The ratios between the individual residual tensile strength and the short-term ultimate, guaranteed, and design tensile strengths.

Figure 5: Residual elastic modulus for the different bars conditioned at elevated temperatures.

Figure 6: Typical cross sections of unconditioned samples (A & B), water conditioned sample (C) and alkaline solution-conditioned sample (D).

Figure 7: Different debonding levels of glass fibres from bars conditioned in water and alkaline solution.

Figure 8: FTIR spectra of 16 mm samples conditioned in deionized water and alkaline solution during 2 months at 60°C.

Figure 9: Typical location of failure on the bars at the ends of the reservoir.

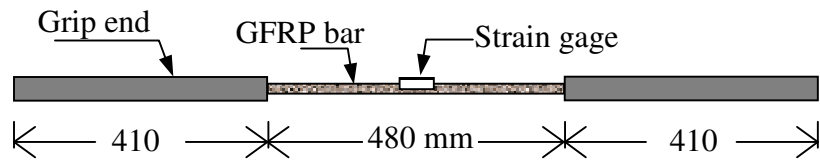


Figure 1a: Details and dimensions of GFRP bar test specimen.

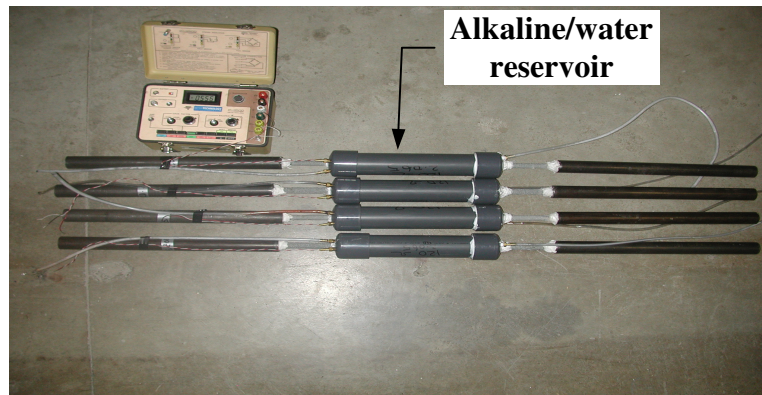


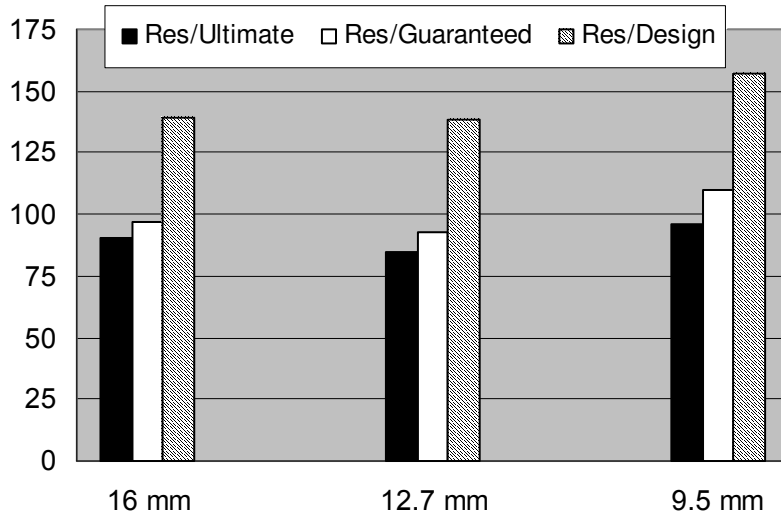
Figure 1b: GFRP test specimens ready for installation.



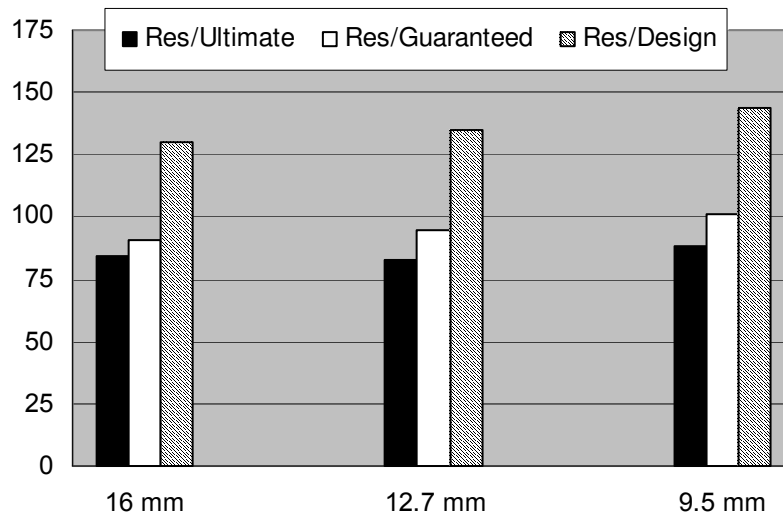
Figure 2: Sustained tensile force applied on GFRP bars through load-magnifying frames.



Figure 3: Steel frames positioned inside the heating tunnel.



in deionized water



in alkaline solution

Figure 4: The ratios between the individual residual tensile strength and the short-term ultimate, guaranteed, and design tensile strengths.

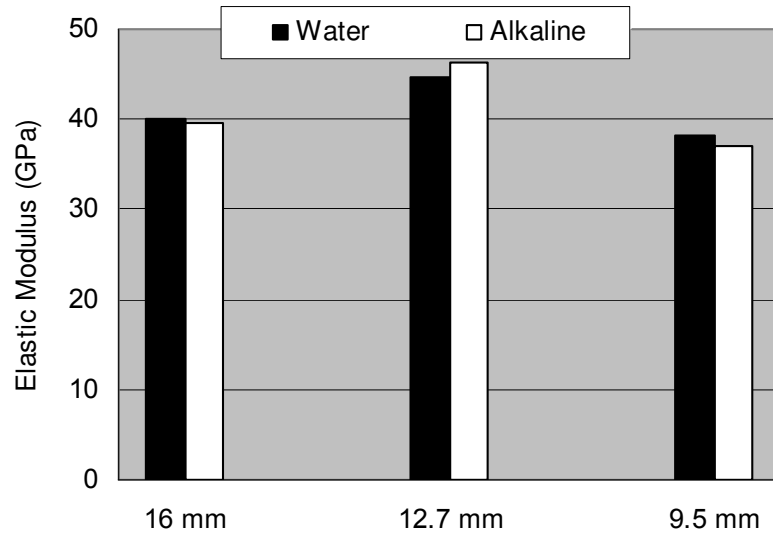
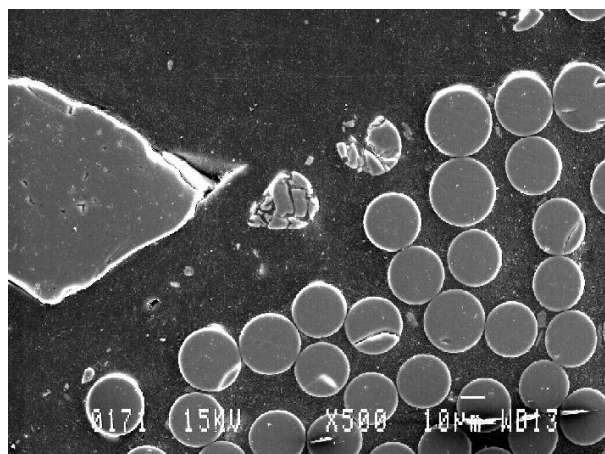
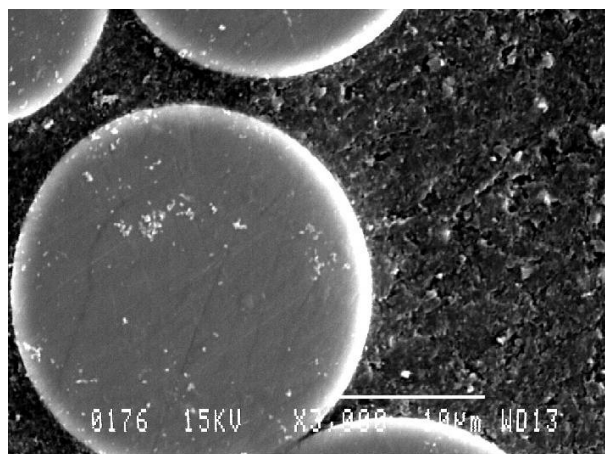


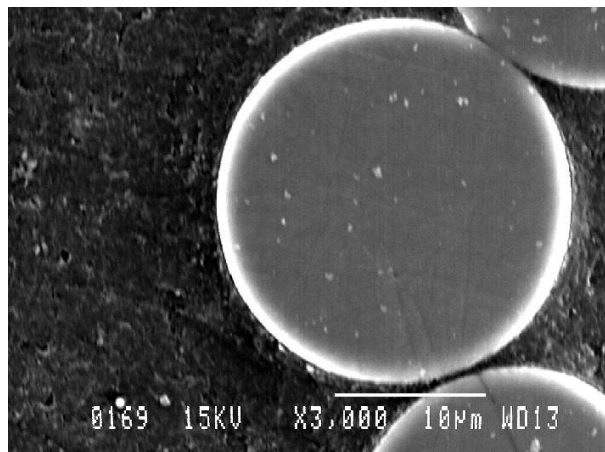
Figure 5: Residual elastic modulus for the different GFRP bars conditioned at elevated temperatures



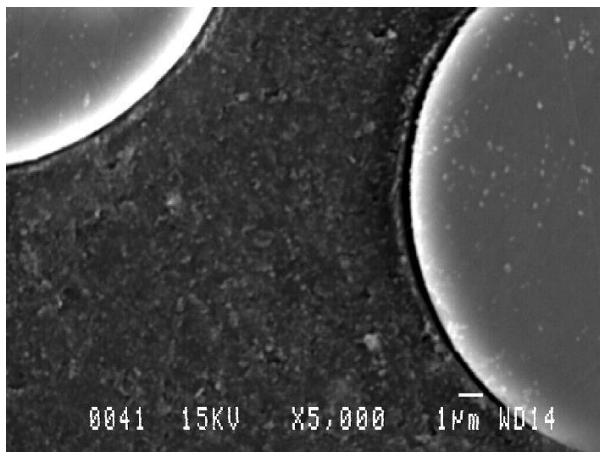
A



B

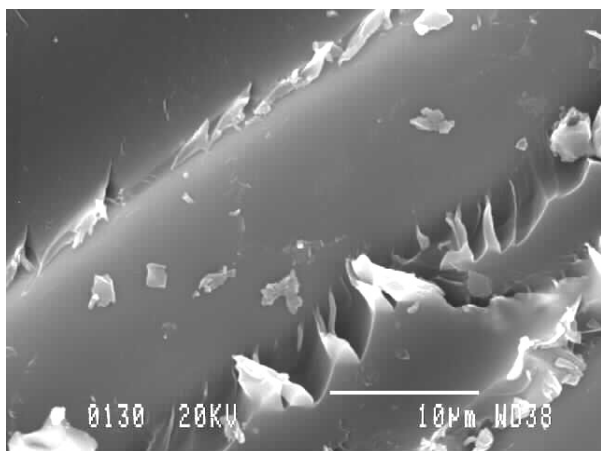


C

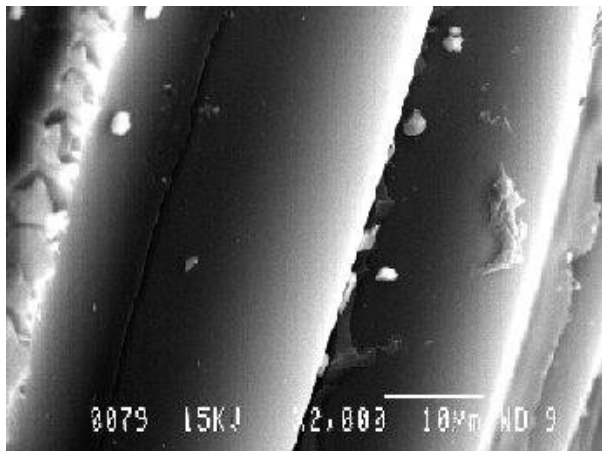


D

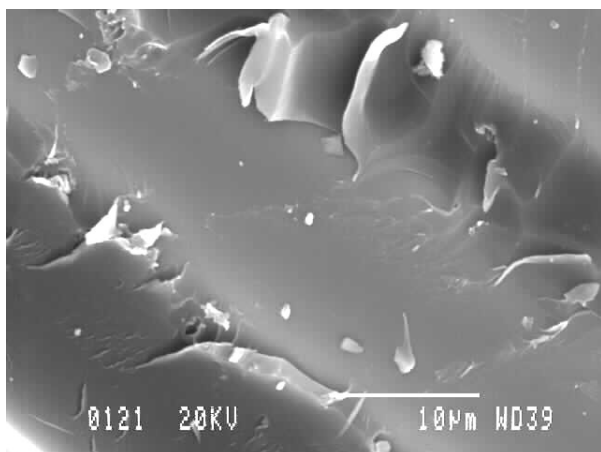
Figure 6: Typical cross sections of unconditioned samples (A & B), water conditioned sample (C) and alkaline solution-conditioned sample (D).



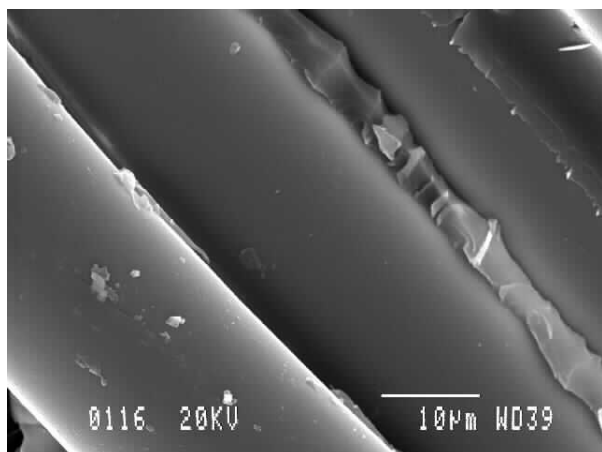
A



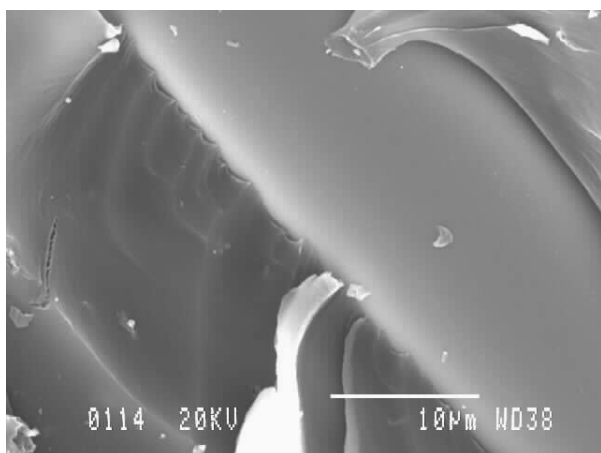
D



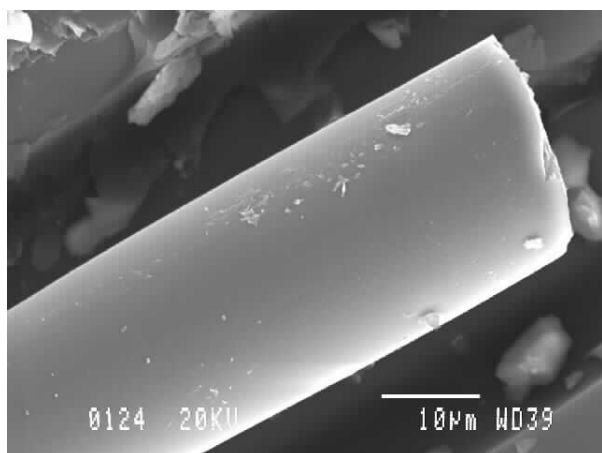
B



E



C



F

Figure 7: Different debonding levels of glass fibres from bars conditioned in water and alkaline solution.

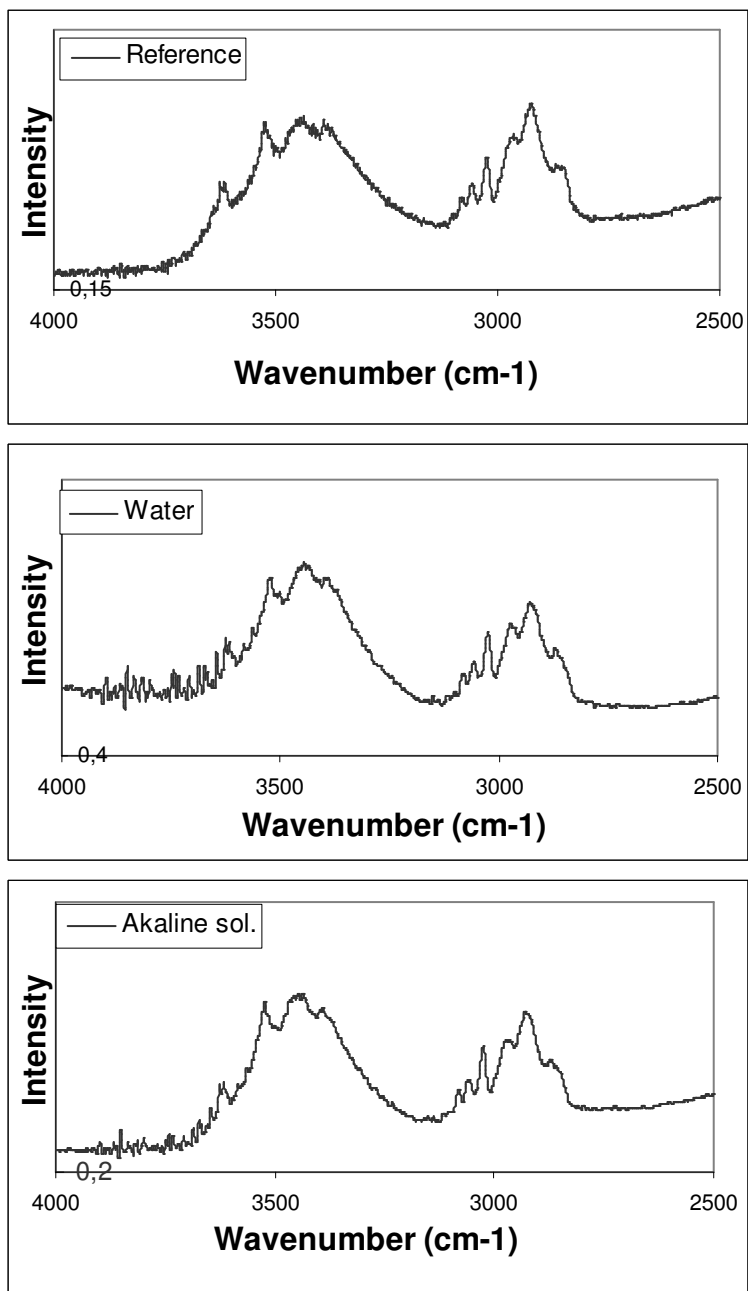


Figure 8: FTIR spectra of 16 mm samples conditioned in deionized water and alkaline solution for 2 months at 60°C.

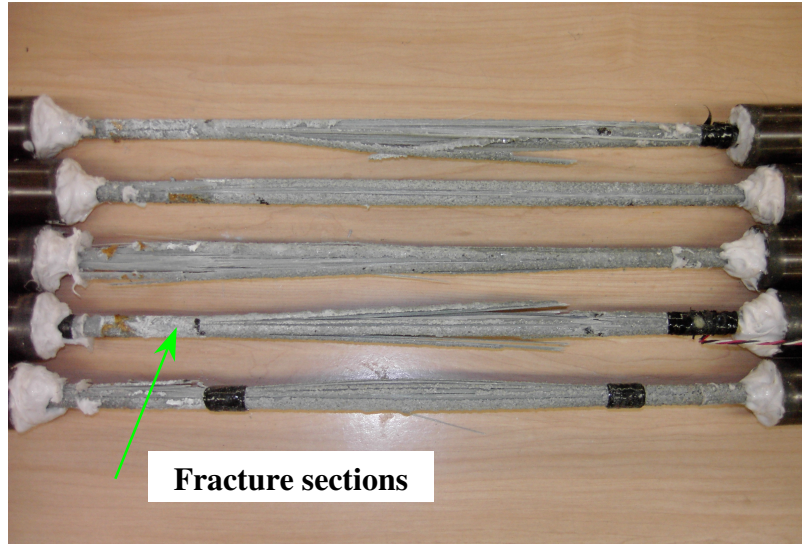


Figure 9: Typical location of failure on the bars at the ends of the reservoir.